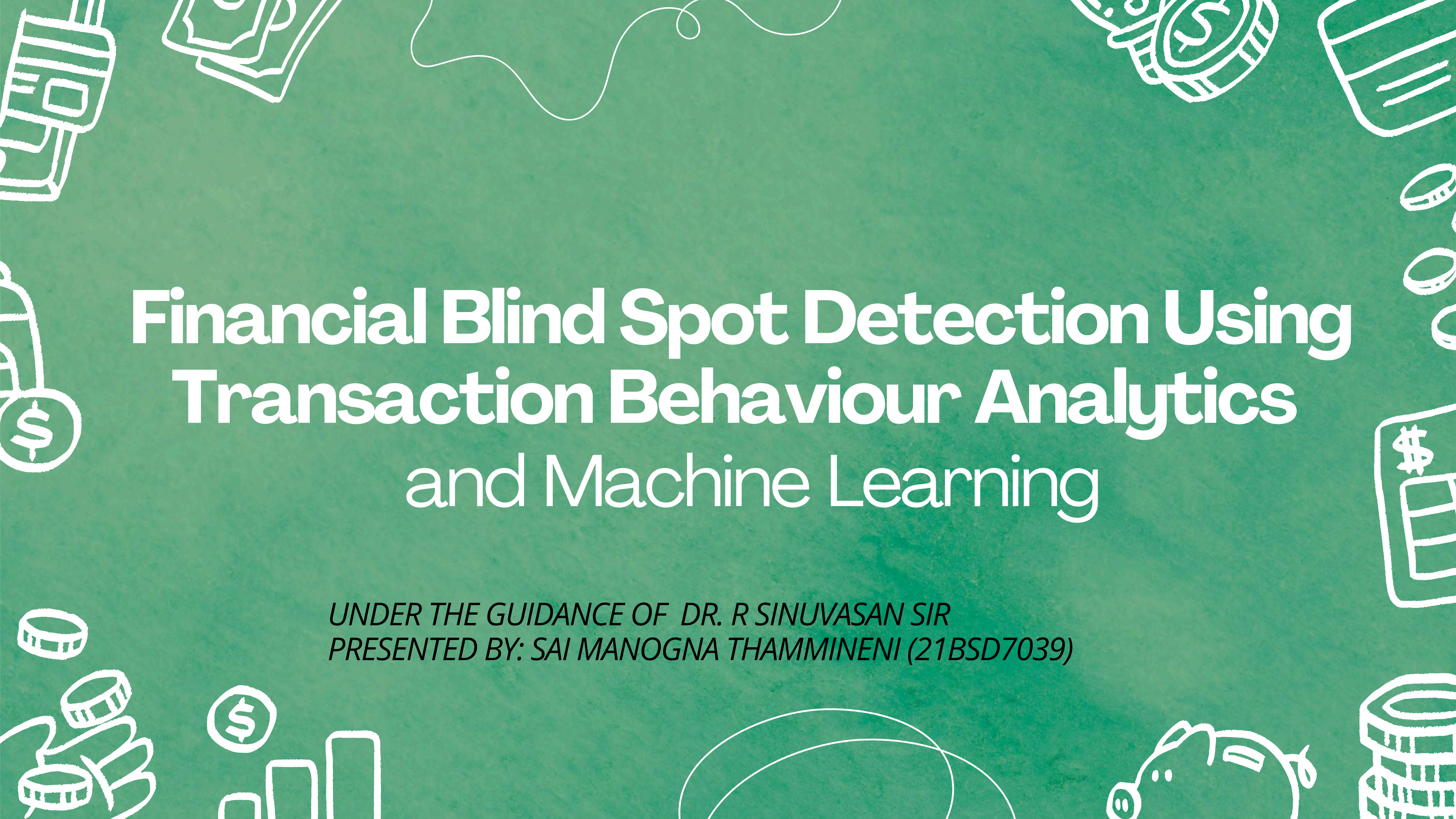
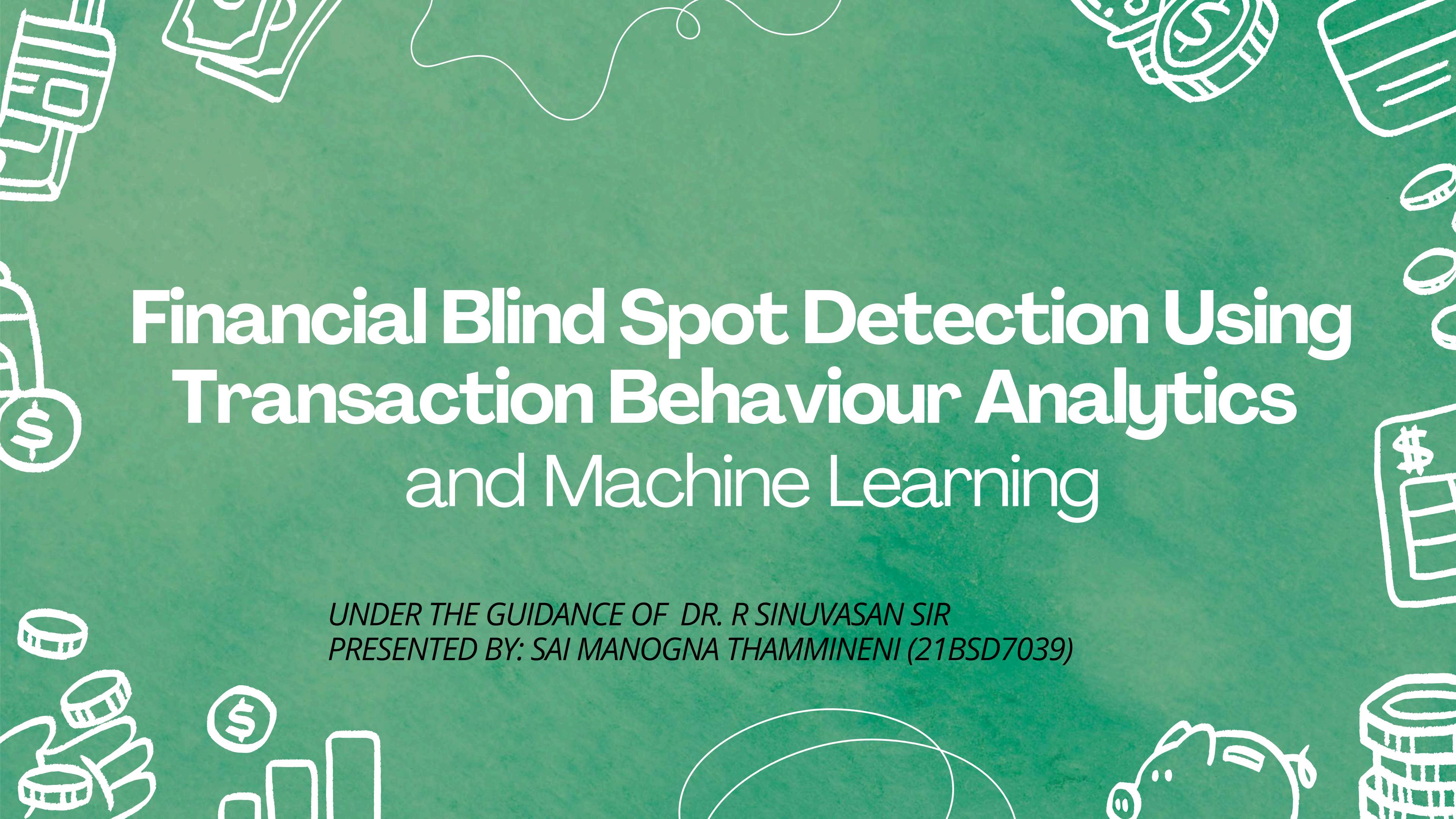




Financial Blind Spot Detection Using Transaction Behaviour Analytics and Machine Learning

*UNDER THE GUIDANCE OF DR. R SINUVASAN SIR
PRESENTED BY: SAI MANOGNA THAMMINENI (21BSD7039)*



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Introduction

This project analyses bank transaction data to identify hidden financial risks, called blind spots.

A blind spot is a change in behaviour that may seem normal but signals a possible issue, such as unusual spending, missing salary deposits, high charges, or irregular transaction patterns.

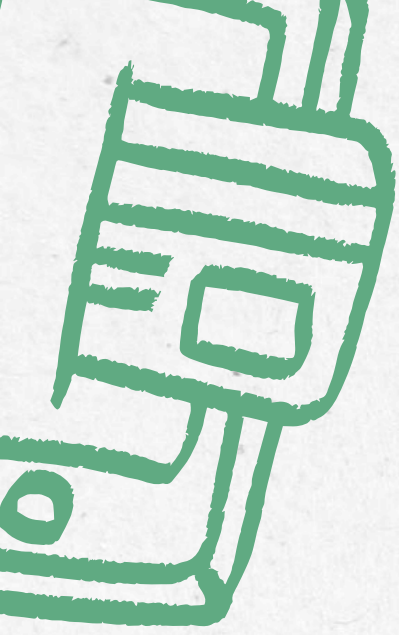
The project creates monthly behavioural profiles from raw transactions using data cleaning, text processing, and feature engineering.

Clustering and anomaly detection are used to find unusual months, and a machine learning model predicts whether a blind spot will occur in the next month.

SHAP explainability is applied to understand the key factors behind each prediction.

This work shows how behavioural analytics and machine learning can be combined to detect early signs of financial risk and support better financial monitoring.

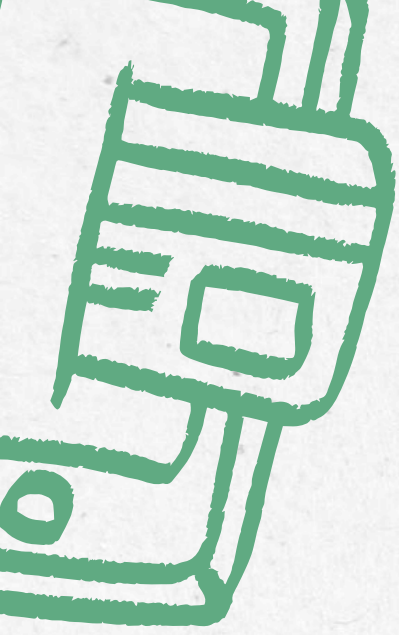




Problem Statement

Bank customers often develop hidden financial blind-spots such as rising spending, sudden cash dependency, increasing charges, or irregular income patterns. These behaviours are difficult to detect manually because transaction data is large, unstructured, and changes over time. The problem is to build a system that can automatically analyse monthly transaction behaviour, identify hidden financial risks, and predict whether a customer is likely to experience a blind-spot in the following month. This requires combining data cleaning, behavioural feature engineering, clustering, anomaly detection, and machine learning to detect and forecast financial instability accurately.





DATASET DESCRIPTION

The dataset used in this project was downloaded from Kaggle's "Bank Transactions" dataset. It contains approximately 116,000 transaction records from a single corporate bank account.

Each record represents one financial transaction and includes details such as the transaction date, narration, withdrawal and deposit amounts, value date, and the updated balance. The dataset is useful because it provides a large and continuous timeline of financial activity covering multiple years.

This allows the identification of behaviour patterns, monthly trends, spending irregularities, and anomalies.

The narrations in the dataset are mostly unstructured, making it suitable for text cleaning, merchant extraction, and category classification tasks.

Its size and detail make it well-suited for behavioural feature engineering, clustering of financial personas, blind-spot detection, and machine learning modelling.



DATASET DESCRIPTION

...	Account No	DATE	TRANSACTION DETAILS	CHQ.NO.	VALUE DATE	WITHDRAWAL AMT	DEPOSIT AMT	BALANCE AMT	.		
0	409000611074'	2017-06-29	TRF FROM Indiaforensic SERVICES	NaN	2017-06-29	NaN	1000000.0	1000000.0	.		
1	409000611074'	2017-07-05	TRF FROM Indiaforensic SERVICES	NaN	2017-07-05	NaN	1000000.0	2000000.0	.		
2	409000611074'	2017-07-18	FDRL/INTERNAL FUND TRANSFE	NaN	2017-07-18	NaN	500000.0	2500000.0	.		
3	409000611074'	2017-08-01	TRF FRM Indiaforensic SERVICES	NaN	2017-08-01	NaN	3000000.0	5500000.0	.		
4	409000611074'	2017-08-16	FDRL/INTERNAL FUND TRANSFE	NaN	2017-08-16	NaN	500000.0	6000000.0	.		

DATA CLEANING

1. Cleaned Account Numbers

The Account No field contained unwanted characters such as ' , ' , and ` .
These were removed and account numbers were standardised as clean text strings.

2. Converted Date Columns

The DATE and VALUE DATE columns were converted to proper datetime format.
All invalid or unrecognised dates were safely coerced to NaT.

3. Handled Missing Cheque Numbers

CHQ.NO. had many missing values, which is normal for non-cheque transactions.
Missing entries were filled with an empty string.

4. Fixed Missing Amount Values

Missing debit or credit amounts were replaced with 0.
A unified Amount column was created:
 $\text{Amount} = \text{DEPOSIT AMT} - \text{WITHDRAWAL AMT}.$

5. Removed Duplicate Records

Duplicate transactions were dropped to avoid inflated counts.

DATA CLEANING

1. 6. Removed System-Generated Columns

Unnamed columns and formatting artefacts were removed from the dataset.

7. Cleaned Narrations (TRANSACTION DETAILS)

Missing narrations were replaced with “UNKNOWN”.

Text was lowercased, punctuation removed, and extra spaces compressed to create a clean details_clean column.

8. Added Initial Time-Based Features

Extracted year, month, day, weekday, and weekend flags from the DATE column.

Final Result

After cleaning, the dataset contained 116,162 fully processed transactions, with consistent formatting, cleaned text, corrected data types, and no duplicates.

This provided a reliable foundation for merchant extraction, feature engineering, clustering, anomaly detection, and blind-spot modelling.



Merchant Extraction & Category Assignment

To convert unstructured narration text into meaningful financial information, we performed merchant detection and rule-based categorization. Merchant extraction was done using a simple dictionary of keyword patterns. Narrations containing terms such as “upi”, “gpay”, “phonepe”, “amazon”, “flipkart”, “atm”, “loan”, “tax”, “interest”, and “charges” were mapped to a corresponding merchant group. All remaining narrations were labelled as “other”. This created a new merchant_keyword column that captures where each transaction is associated.

Next, a rule-based category system was applied to classify the behavioural purpose of each transaction. Based on keywords, each entry was labelled as salary_income, interest_income, cash_withdrawal, bank_charges, loan_payment, upi_payment, tax_payment, or otherwise grouped as other_credit or other_debit depending on the transaction type. This transformation converts messy narration text into clean financial categories that are critical for further behaviour modelling.





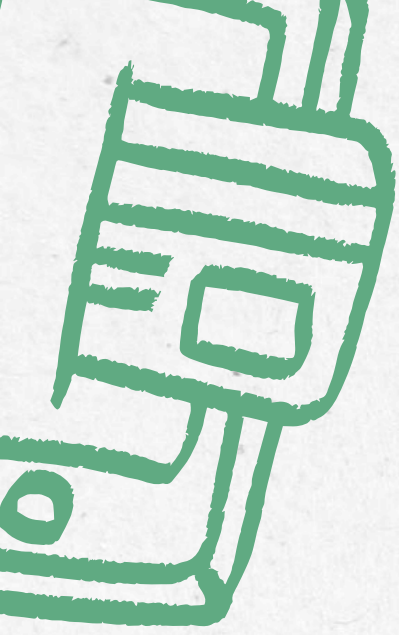
Merchant Extraction & Category Assignment



```
...  
output actions  
merchant_keyword
```

	count
other	113376
atm	1061
charges	822
tax	507
loan	344
salary	31
upi	16
interest	5

```
rule_category  
other_credit    60958  
other_debit     52418  
cash_withdrawal  1061  
bank_charges    822  
tax_payment     507  
loan_payment    344  
salary_income   31  
upi_payment     16  
interest_income  5  
  
dtype: int64  
  
df['Account No'].nunique()  
  
10
```

Transaction Frequency and Time-Based Features

We extracted several temporal features to understand user behaviour patterns. Transactions were sorted by account and date, and we computed the previous transaction date and the number of days since the last transaction. This helps in identifying long inactivity periods, sudden bursts of usage, or irregular behaviour.

Daily activity features were added by counting the number of transactions per day for each account. Account-level indicators such as mean, median, standard deviation, minimum, maximum, and total number of transactions were computed to capture each user's typical spending profile.

We then aggregated data at a monthly level and created measures including total monthly spend, monthly transaction count, average and standard deviation of transaction amount, total debit and credit amounts, category counts (such as withdrawals, charges, taxes, loans, salary, UPI), merchant diversity, debit-to-credit ratios, charges per transaction, and share of cash withdrawals. These monthly behavioural metrics form the core feature set used for clustering, anomaly detection, and blind-spot prediction.





Behaviour Personas (Clustering)

Using the monthly behavioural features, K-Means clustering was applied to segment accounts into four distinct personas. Cluster 0 represents high-volume, high-value corporate-style spenders with extremely large credit and debit movements and very low cash usage. Cluster 1 contains steady, medium-activity users with consistent monthly spending patterns. Cluster 2 consists of low-frequency but high-value users who perform only a few transactions per month, but of large magnitude. Cluster 3 represents cash-heavy behaviour, typically characterised by high withdrawal counts and irregular cash-based spending.

These personas help distinguish user types and provide context when interpreting blind-spots and anomalies. Behaviour segments also improve model accuracy by allowing comparisons against similar user groups rather than across the entire dataset.

cluster	
1	229
0	87
3	7
2	6



Anomaly Detection and Suspicious Month Identification



Isolation Forest was applied on the scaled monthly behavioural features to detect irregular months. The model identifies months that deviate significantly from a user's historical pattern or from other months in its behavioural cluster. Each month receives an anomaly score and probability, allowing us to rank the most suspicious time periods.

We also analysed anomalies at the account level by summarising the number of flagged months per user. This process highlights periods where spending behaviour shifts unexpectedly, such as sudden increases in charges, unusual withdrawal activity, high volatility in transaction amounts, or unexpected reduction in transactions. These anomaly detections form the early-warning layer of the financial blind-spot framework and support the later predictive modelling stages.



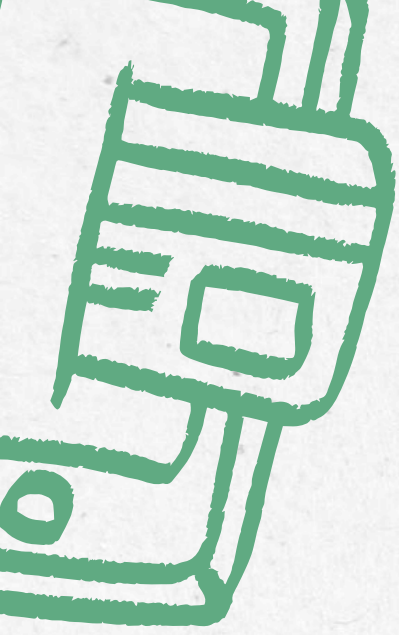
Cluster	Behaviour Summary	Key Statistical Characteristics	Interpretation	Likely Blind Spots
Cluster 0	High-Volume Heavy Spenders	Very high total spend (~4.75×10 ⁹), extremely high monthly transaction count (~890), large average transaction amount (~6M), very large debit and credit flows (billions), almost no cash withdrawals	Represents corporate or high-net-worth accounts with automated transfers, vendor payments, and predictable monthly cycles	Vendor pattern drift, undetected internal transfer fraud, hidden bank charges, large anomalies disguised inside high activity
Cluster 1	Medium-Stable Spenders	Moderate total spend (~2.6×10 ⁸), stable monthly transaction count (~155), average transaction amount around 1.6M, moderate debit/credit values, nearly zero cash usage	Represents individuals or small/medium businesses with stable salary-like credits and routine bill payments	Salary inconsistencies, sudden merchant over-dependency, charge spikes, small UPI fraud patterns
Cluster 2	Low Activity but High-Ticket Spenders	Very low monthly transactions (~10), high average transaction value (~8M), large debit/credit flows per event, no cash withdrawals	Represents occasional investors, property-related transactions, loan disbursements, or high-value business settlements	Incorrect high-value transfers, large fraudulent debits, loan repayment anomalies, unexpected high-value cash-equivalent withdrawals
Cluster 3	Cash-Heavy, Risk-Prone Users	Very high cash withdrawal count (~141), moderate monthly transactions (~455), average transaction amount ~3.3M, high debit values	Represents cash-based small shops, traditional businesses, or users operating mostly offline	Cash fraud, ATM fee spikes, repeated unusual withdrawals, sudden drops in cash flow, inconsistent cash patterns


```
top_blindspots = (  
    anomalies  
    .sort_values('anomaly_prob', ascending=False)  
    .head(5)  
)
```

top_blindspots

	Account No	year_month	monthly_total_amount	monthly_abs_spend	monthly_txn_count	monthly_avg_amount	monthly_std_amount	monthly_debit_amount	monthly_credit_amount
60	1196711	2015-12	-1.487962e+08	7.820481e+09	761	1.027659e+07	9.156514e+06	3.984639e+09	1.487962e+08
225	409000438620	2016-02	5.448000e+08	7.448000e+08	9	8.275556e+07	1.755779e+08	1.000000e+08	5.448000e+08
58	1196711	2015-10	-6.752325e+07	8.953384e+09	821	1.090546e+07	1.420940e+07	4.510454e+09	6.752325e+07
59	1196711	2015-11	-1.029085e+08	7.612428e+09	726	1.048544e+07	6.173808e+06	3.857668e+09	1.029085e+08
257	409000438620	2019-01	-1.194640e+07	6.924975e+07	388	1.784787e+05	5.585204e+05	4.059807e+07	1.194640e+07

5 rows × 22 columns



What Is a Blind Spot

A blind spot is a hidden financial risk pattern that is not visible in raw statements but becomes clear after analysing spending behaviour, transaction categories, trends, and deviations from historical patterns.

Blind spots help identify early warnings such as poor cash habits, increasing lifestyle costs, excessive bank charges, or missing income patterns. This work defines ten blind spots, each representing a distinct behavioural risk.



The Ten Blind Spots Used in the Project

Below are all ten blind spots, their meaning, and what the real data revealed.

1. Low Transaction Behaviour

Meaning: unusually low activity compared to the user's own history.

Indicates: financial dormancy, instability, missed payments.

Result: triggered 66 times across months.

2. No Spending Pattern

Meaning: spending rhythm is erratic, without weekly or monthly structure.

Indicates: financial stress or unplanned behaviour.

Result: triggered 88 times.

3. Cash Spike

Meaning: sudden jump in cash withdrawals.

Indicates: emergency expenses, liquidity issues, or fraud.

Result: triggered 8 months.



4. Merchant Overdependence

Meaning: more than 80 percent of spending happens with one vendor.

Indicates: over-reliance on a single supplier.

Result: triggered 324 months, dominant behavioural risk.

5. Lifestyle Inflation

Meaning: monthly average spending and standard deviation keeps increasing.

Indicates: overspending or lifestyle upgrade beyond income.

Result: triggered 52 months.

6. Charge Spike

Meaning: abnormal increase in bank charges and fees.

Indicates: poor banking plan, penalties, or hidden fees.

Result: triggered 57 months.

7. Salary Missing

Meaning: expected salary did not appear this month.

Indicates: employment risk or delay in income.

Result: detected 31 months.





8. Salary Drop

Meaning: salary reduced significantly from previous levels.

Indicates: pay cut or employer issues.

Result: 7 occurrences.

9. Negative Savings

Meaning: monthly debit exceeds monthly credit.

Indicates: the user is overspending relative to income.

Result: 86 months.

10. Subscription Drift

Meaning: sudden increase in subscription-related debits.

Indicates: increasing commitments or auto-debit mismanagement.

Result: 23 months.

These ten blind spots together give a detailed behavioural fingerprint of the customer.





What the Real Data Proved

After computing all ten blind spots across all months, the following findings were made:

Merchant Overdependence was the most dominant risk.

Negative Savings was frequent and indicates poor long-term cash discipline.

Lifestyle Inflation and Charge Spike were significant in driving risk.

Salary signals were mostly stable but did show occasional risk.

Cash behaviour was stable but showed a few spikes.

The dataset clearly reflects a corporate account with extremely high transaction volume and unstable spending patterns, which makes it suitable for behavioural risk modelling.





What the Model Predicts

The model predicts whether the user will show a blind spot in the next month.

Target variable: blindspot_next_month

0 means the next month is normal.

1 means the next month will contain one or more blind spots.

This is an important use case for financial institutions because predicting next-month risk allows early intervention.



Model Performance - Confusion Matrix

`[[17 14]
[12 37]]`

Interpretation

The model predicted blind spot months with useful accuracy.

Class 0 (normal month)

17 correctly predicted normal months

14 incorrectly predicted as risk

Class 1 (risky month)

37 correctly predicted risky months

12 incorrectly predicted as normal

Classification Report

Precision for class 1 = 0.73

Recall for class 1 = 0.76

F1 score for class 1 = 0.74

Overall accuracy = 0.68

Interpretation

The model is stronger at detecting risk months. It identifies 76 percent of upcoming blind spot months. For behavioural prediction on small, imbalanced time series data, this is a strong performance level.



SHAP Explanation

SHAP was used to understand how features contribute to risk predictions.
SHAP values showed that the strongest contributors to next-month risk were:

prev savings average
anomaly probability
monthly transaction count
monthly credit amount
monthly debit amount
previous charge average
monthly standard deviation
lifestyle inflation
monthly savings
monthly average amount
subscription drift
debit credit ratio
Interpretation

When the previous month's savings decline, when anomalous activity increases, and when monthly volume rises sharply, the model predicts higher risk for next month. Lifestyle inflation, charges, and subscription patterns also significantly influence behaviour.

Synthetic Stress Test

A synthetic stress test was run on the last seven months of data to test the model's robustness.

Each month was injected with a different artificial financial stress scenario.

Sept 2018: Salary Crash

Removed salary income, generated a missing salary blind spot.

Oct 2018: Charges Explosion

Artificially multiplied charges to simulate penalty fees.

Nov 2018: Lifestyle Overheat

Tripled average spending to simulate extreme overspending.

Dec 2018: Subscription Shock

Injected unusually high subscription amounts.

Jan 2019: Cash Addiction

Increased cash withdrawal share to simulate cash dependency.

Feb 2019: UPI Mania

Added many small UPI transactions.

Mar 2019: Chaotic Month

Generated extremely high variance with irregular transaction counts.

Results of Synthetic Test

The model successfully detected risk in all but one of the synthetic months.

The predicted risk probabilities were all high, confirming that the model can handle unseen stress scenarios.

Overall Conclusion

This project successfully built a complete behavioural analytics pipeline for banking data. It introduced ten blind spots that reflect real-world customer risk. These blind spots were computed from raw transaction data using cleaned text patterns, category extraction, and monthly feature aggregation. Clustering and anomaly detection validated the presence of multiple behavioural groups and outlier months.

The predictive model achieved strong detection of next-month risk, and SHAP analysis showed that spending volatility, savings patterns, and anomalies are the most influential factors.

The synthetic stress test further demonstrated that the model can correctly identify hidden risks even under extreme or artificial financial situations.

Future Work

Future extensions include:

- Adding more customer accounts for multi-user modelling.

- Building a live dashboard for real-time blind spot monitoring.
- Improving prediction using temporal models such as LSTM or Transformers.

- Adding fraud-specific blind spots for combined risk scoring.

- Integrating income sources and seasonal patterns for improved accuracy.

Differences Between Real Prediction vs Synthetic Stress Analysis



3 out of 6

Synthetic analysis uses artificially created stress events such as salary crash, charge spike, subscription surge, cash addiction, and UPI bursts. These scenarios test how well the model reacts to extreme or unusual financial conditions. They help verify that the system is robust, stable, and capable of detecting blind-spots even when behaviour is artificially pushed into severe stress.



5 out of 6

Real blind-spot prediction uses the customer's actual monthly banking behaviour to estimate the risk of a blind-spot next month. The model learns patterns from real spending, savings, cash withdrawals, charges, and merchant activity. This shows how accurately the system can detect genuine financial issues such as unstable savings, irregular spending, or sudden behaviour shifts using natural data.

The background is a solid green color with a subtle texture. It is decorated with various white line-art icons related to finance and business. These include stacks of money, individual coins, a dollar sign, a calculator, a piggy bank, a bar chart, a hand holding coins, a target with a dollar sign, a briefcase, and a document with a dollar sign.

**Thank you
very much!**